Discrete Probability Distributions

Dr. Skip Crooker MBA 775 University of Nevada, Las Vegas

Thursday, October 22, 2026

Table of contents

- Objectives . . . 2
- Random Variables . . . 2
- Discrete Probability Distributions . . . 3
  - Where a distribution comes from . . . 3
  - Application: defective units . . . 3
  - Is it a distribution? . . . 4
  - The mean of a discrete probability distribution . . . 5
  - The variance and standard deviation . . . 6
  - Comparing two distributions . . . 7
- The Expected Monetary Value . . . 8
  - Application: RAD Construction . . . 9
  - What if the contract changes? . . . 9
  - Choosing among alternatives . . . 9
- The Binomial Distribution . . . 10
  - The characteristics of a binomial experiment . . . 10
  - Building the formula from Chapter 4 . . . 10
  - The mean and standard deviation of a binomial distribution . . . 12
  - Application: Corporate Headquarters . . . 12
  - Translating the question . . . 13
  - Reading the output . . . 14
  - The shape of a binomial distribution . . . 15
  - Does the binomial describe Nevada? . . . 15
  - Using software to calculate binomial probabilities . . . 17
- The Poisson Distribution . . . 17
  - The characteristics of a Poisson process . . . 17
  - Using Poisson probability distributions . . . 17
  - Application: Corporate Headquarters, one more hour . . . 18
  - Changing the interval . . . 19
  - Using the Poisson distribution to approximate the binomial . . . 20
  - Using software to calculate Poisson probabilities . . . 21
- The Hypergeometric Distribution . . . 21
  - The auditor, continued . . . 21
  - Application: selecting store managers . . . 22
  - Application: was the selection random? . . . 22
  - Using software to calculate hypergeometric probabilities . . . 23
- Which Distribution Is It? . . . 23
- Applications . . . 24
  - The help desk . . . 24

The supplier's claim . . . 25
Laboratory 5 . . . 26
Path 1 — upload the files . . . 26
Path 2 — let Claude fetch the files . . . 27

Objectives

- Identify the rules of a discrete probability distribution.
- Calculate and interpret the mean and standard deviation of a discrete distribution.
- Use expected monetary value to compare alternatives, and say what it leaves out.
- Understand the characteristics of a binomial experiment, and recognize when data violate them.
- Calculate binomial probabilities, and interpret the mean and standard deviation of a binomial distribution.
- Understand the characteristics of a Poisson process, and calculate Poisson probabilities over any interval.
- Approximate binomial probabilities using the Poisson distribution.
- Calculate hypergeometric probabilities.
- Translate a question asked in English ("fewer than," "at least") into the right cell of a probability table.
- Use a probability to judge whether a claim is believable.

In Chapters 2 and 3 we described a sample we already had. In Chapter 4 we put a probability on a single event. This chapter puts a probability on *every* value a count can take, all at once. That complete list is a **probability distribution**, and it is what I have been calling a **data generating mechanism**: the rules that determine the characteristics we find in any sample drawn from it.

The reason to care is that a manager almost never gets to see the mechanism. She sees one sample: five big spenders out of fifty customers, four defects in a shipment of twenty. The question she needs answered is whether that sample is the kind of thing the mechanism she *believes in* would produce. This chapter is where we first learn to answer it, and Chapter 9 is where we make the answer formal.

You will not be asked to grind through these formulas by hand on an exam. You will be handed output like the tables in this note and asked which number answers the question, and what a manager should conclude from it. So read the tables closely. They are the skill.

Random Variables

A **random variable** is a numerical outcome of an experiment. Its value cannot be known with certainty before the experiment is run, and is known exactly afterwards. By convention we write it x.

Discrete random variables have outcomes that typically take on whole numbers as a result of conducting an experiment. They are usually *counted*.

Continuous random variables have outcomes that can take on any numerical value as a result of conducting an experiment. They are usually *measured*.

Application: a discrete random variable

If we draw n units at random from a production line at our manufacturing facility, how many of them will be defective? The answer is a whole number. The count of defective units must be in the set $\{0, 1, 2, ..., n\}$.

Application: a continuous random variable

How much time elapses before a product made at our facility fails? We can measure time in increments as small as we like, so the answer may be any real number in $(0, \infty)$. It may be 300.12311 hours.

The acid test is to ask how many values are possible inside an interval. A customer rating from 1 to 5 has five. A waiting time between 0 and 5 minutes has infinitely many, limited only by the precision of the clock.

There is a practical exception you met in Chapter 2. Income is technically discrete, since we count dollars and cents, but it has so many possible values that treating it as continuous is the sensible choice. Continuous random variables are Chapter 6.

Discrete Probability Distributions

A listing of all the possible outcomes of an experiment for a discrete random variable, along with the probability of each outcome, is called a **discrete probability distribution**.

A discrete probability distribution meets three conditions:

1. The outcomes are **mutually exclusive**. The value of the random variable cannot fall into more than one row of the table.
2. Every probability lies between 0 and 1: $0 \le P(x) \le 1$ for every x.
3. The probabilities sum to 1: $\sum_{i=1}^{n} P(x_i) = 1$, where n is the number of possible outcomes.

These are Rules 3 and 4 from Chapter 4, applied to a whole table at once.

Where a distribution comes from

The three kinds of probability from Chapter 4 give three kinds of distribution, and it is worth knowing which one you have been handed.

- **Empirical.** Take a frequency table from Chapter 2, divide each count by the total, and the relative frequency column *is* a probability distribution. It is as good as the data behind it.
- **Classical, from a model.** If you can describe the mechanism (fifty independent customers, each with the same chance of spending $100), the distribution follows from a formula. The binomial, Poisson, and hypergeometric distributions later in this note are all of this kind.
- **Subjective.** A project manager's judgment that a bridge will finish early, on time, or late with probabilities 25%, 60%, and 15% is a distribution too.

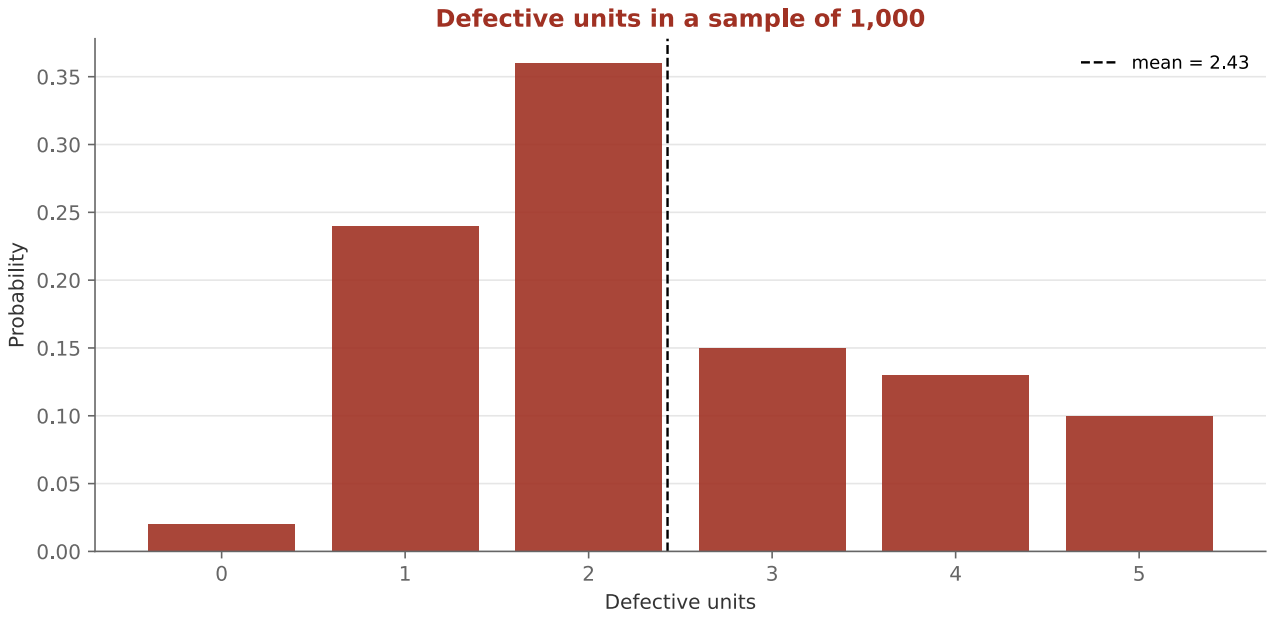
Application: defective units

Given the design specifications for our production lines, the number of defective units in a random sample of 1,000 from the line follows the distribution below.

defective units	P(x)
0	0.02
1	0.24
2	0.36
3	0.15
4	0.13
5	0.10

```
every P(x) in [0, 1]     True
sum of P(x)               1.0
```

```
sums to 1              True
verdict                valid
dtype: object
```



Six or more defective units has probability zero under these specifications, so those rows are left off the table.

Is it a distribution?

A bakery's analyst offers five candidate distributions, A through E, for the daily demand for a specialty loaf. Which are valid discrete probability distributions? For each that is not, say which rule it breaks.

daily demand	A	B	C	D	E
0	0.1	0.1	0.3	0.15	0.0
1	0.2	0.2	0.3	0.25	0.0
2	0.4	0.4	0.5	0.30	1.0
3	0.2	0.2	−0.1	0.20	0.0
4	0.1	0.2	0.0	0.05	0.0

i Solution

	every P(x) in [0, 1]	sum of P(x)	sums to 1	verdict
A	True	1.0	True	valid
B	True	1.1	False	NOT valid: the column sums to 1.1, not 1
C	False	1.0	True	NOT valid: −0.1 is outside [0, 1]
D	True	0.95	False	NOT valid: the column sums to 0.95, not 1
E	True	1.0	True	valid

A and E are valid. B sums to more than 1 and D to less than 1. C sums to 1, which is why it is the trap: one of its entries is negative, and no probability can be.

E looks odd but breaks no rule. It says demand is 2 loaves every day with certainty. Hold on to it. It is the distribution with no variability at all, and we will use it to check that the variance formula behaves.

The mean of a discrete probability distribution

The **mean** of a discrete probability distribution is

$$\mu = \sum_{i=1}^{n} x_i \, P(x_i),$$

where x_i is the value of the random variable for the ith outcome, $P(x_i)$ is the probability that the ith outcome occurs, and n is the number of outcomes in the distribution.

Expected value is another term for the mean of a probability distribution, written $E(x) = \mu$.

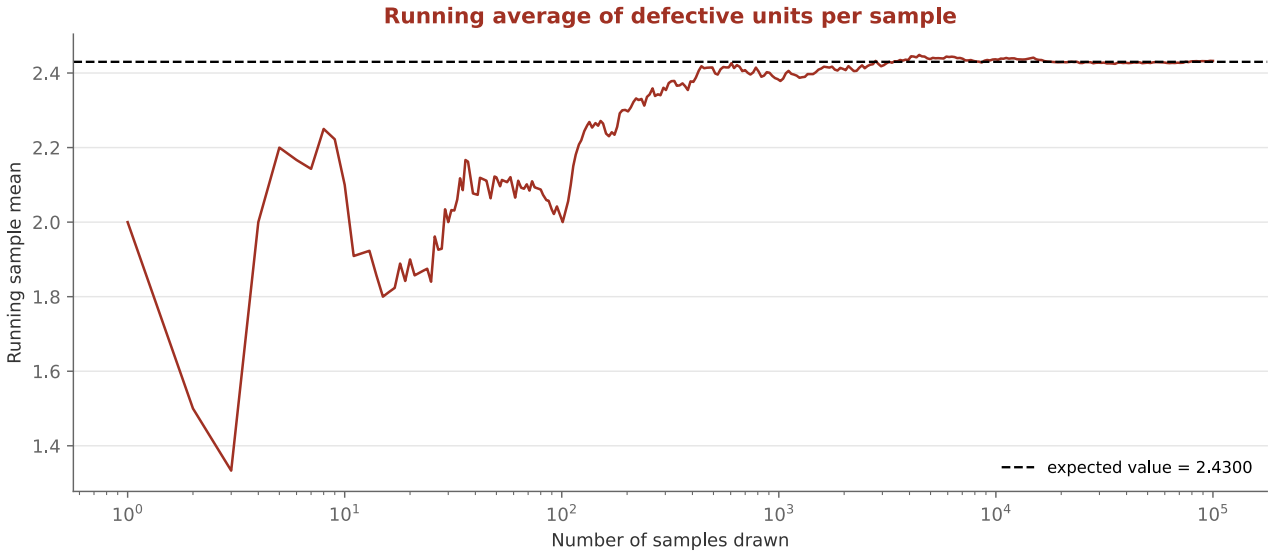
The mean is a weighted average of the outcomes, with the probabilities as the weights. The table below works it out in columns, the way you would on paper. The `x P(x)` column sums to the mean.

defective units	P(x)	x P(x)	x - mu	(x - mu)^2	(x - mu)^2 P(x)	x^2 P(x)
0	0.02	0.00	−2.43	5.9049	0.1181	0.00
1	0.24	0.24	−1.43	2.0449	0.4908	0.24
2	0.36	0.72	−0.43	0.1849	0.0666	1.44
3	0.15	0.45	0.57	0.3249	0.0487	1.35
4	0.13	0.52	1.57	2.4649	0.3204	2.08
5	0.10	0.50	2.57	6.6049	0.6605	2.50
TOTAL	1.00	2.43			1.7051	7.61

$$\begin{aligned}\mu &= 0(0.02) + 1(0.24) + 2(0.36) + 3(0.15) + 4(0.13) + 5(0.10) \\ &= 0.00 + 0.24 + 0.72 + 0.45 + 0.52 + 0.50 \\ &= 2.43\end{aligned}$$

No sample will ever contain 2.43 defective units. The mean of a distribution does not have to be one of its possible outcomes, and here it is not.

What the number describes is the long run. If we drew sample after sample from this line, the *average* number of defective units per sample would settle on 2.43. That is the law of large numbers from Chapter 4, applied to an average instead of a proportion, and we can watch it happen.



After ten samples the running average is 2.10. After 100,000 it is 2.4324, against an expected value of 2.43. The distribution is the mechanism; the samples are what it generates.

The variance and standard deviation

The **variance** of a discrete probability distribution is

$$\sigma^2 = \sum_{i=1}^{n} (x_i - \mu)^2 \, P(x_i),$$

and the **standard deviation** is its square root, $\sigma = \sqrt{\sigma^2}$.

A shortcut that gives the identical answer with less arithmetic:

$$\sigma^2 = \left(\sum_{i=1}^{n} x_i^2 \, P(x_i) \right) - \mu^2.$$

Both routes are in the table above. The `(x - mu)^2 P(x)` column sums to the variance directly, and the `x^2 P(x)` column is the first term of the shortcut.

$$\sigma^2 = 7.6100 - (2.43)^2 = 7.6100 - 5.9049 = 1.7051 \qquad \sigma = \sqrt{1.7051} = 1.3058$$

The variance is 1.7051 *units squared*, which nobody can picture. The standard deviation, 1.31 defective units, is in the units of the data, and it is the one to report. It says a typical sample lands roughly 1.3 units away from the mean of 2.43.

Recall candidate E from the exercise above, where demand is 2 with certainty. Its mean is 2, every deviation $(x - \mu)$ that has any probability is zero, and so its variance is 0.0. A distribution with no spread has no variance. The formula is measuring what we want it to measure.

In Excel, with outcomes in `A2:A7` and probabilities in `B2:B7`, the mean is `=SUMPRODUCT(A2:A7, B2:B7)`. With the mean in cell `D1`, the variance is `=SUMPRODUCT((A2:A7-D1)^2, B2:B7)`.

Comparing two distributions

A retailer collects one-to-five-star ratings from 100 customers at each of two stores, Henderson and Summerlin.

(a) Convert each store's frequencies into a probability distribution.

(b) Calculate the mean and standard deviation of each.

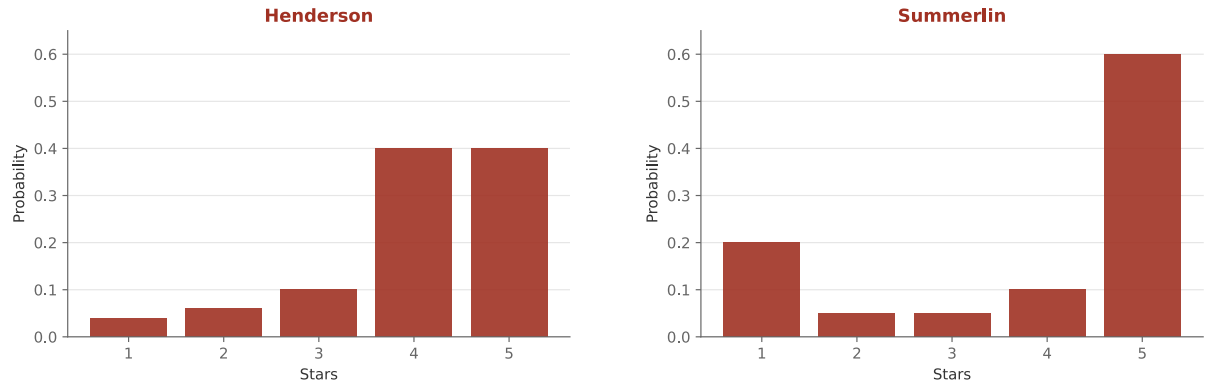
(c) Regional management ranks stores by mean rating. What does that ranking miss here?

stars	Henderson	Summerlin
1	4	20
2	6	5
3	10	5
4	40	10
5	40	60

Solution

stars	Henderson P(x)	Summerlin P(x)
1	0.04	0.20
2	0.06	0.05
3	0.10	0.05
4	0.40	0.10
5	0.40	0.60

	mean	standard deviation
Henderson	4.06	1.047
Summerlin	3.85	1.621



The means are 4.06 and 3.85, close enough that a ranking by mean would call the stores similar. The standard deviations, 1.05 and 1.62, say they are not. Summerlin has *more* five-star customers than Henderson and five times as many one-star customers. It is a store people either love or leave, and a mean cannot show that.

A word of caution about that example. Star ratings and Likert scales ("strongly disagree" to "strongly agree") are **ordinal** data, in the vocabulary of Chapter 1. Nothing guarantees that the distance from two stars to three is the same as from four to five, and a mean quietly assumes it is.

Averaging ratings is nearly universal practice, and we will not pretend otherwise. Just know that when you report a mean rating you have treated ordinal data as interval data, and that some of your audience will notice.

The Expected Monetary Value

The **expected monetary value (EMV)** is the mean of a discrete probability distribution when the random variable is expressed in dollars.

Application: RAD Construction

The RAD Construction Company has been awarded a contract with the State of California to replace a bridge on Pike Creek Road. If the project is completed on time, defined as during the month of September, RAD expects to earn $100,000 in profit. If it is completed before September, the contract pays a $20,000 bonus. If it is completed after September, there is a $10,000 penalty. RAD estimates the probabilities of finishing early, on time, and late at 25%, 60%, and 15%. What is the EMV of the project?

alternative	payoff if early	payoff if on time	payoff if late	EMV	SD	CV
original contract	$120,000	$100,000	$90,000	$103,500	$10,137	0.0979

$$\begin{aligned} EMV &= 0.25(\$120,000) + 0.60(\$100,000) + 0.15(\$90,000) \\ &= \$30,000 + \$60,000 + \$13,500 \\ &= \$103,500 \end{aligned}$$

> RAD will earn $90,000, $100,000, or $120,000. It will never earn $103,500. So what is that number?
>
> It is the long-run average profit *if this same project were undertaken over and over*. The bridge will be built once. The EMV is still the right summary for a firm that bids on many projects, because across a portfolio of them the averages are what show up in the annual results. For a firm betting everything on one project, the EMV is a much weaker guide, and the spread around it matters more.

So we measure the spread. The standard deviation of the payoff is $10,136.57, and the coefficient of variation from Chapter 3, $CV = \sigma/\mu$, is 0.0979.

What if the contract changes?

Suppose the state offers a modified contract: a $23,000 bonus for finishing early, a $15,000 penalty for finishing late, and RAD's probabilities unchanged. Is the modified contract preferred to the original?

alternative	payoff if early	payoff if on time	payoff if late	EMV	SD	CV
original contract	$120,000	$100,000	$90,000	$103,500	$10,137	0.0979
modified contract	$123,000	$100,000	$85,000	$103,500	$12,400	0.1198

Both contracts have an EMV of $103,500. On EMV alone RAD is indifferent. But the modified contract has a standard deviation of $12,399.60 against $10,136.57, and a coefficient of variation of 0.1198 against 0.0979. The same expected money with less risk is the better deal. We would prefer the original contract.

Choosing among alternatives

The most common business use of EMV is choosing how much of something to commit to before you know the demand.

> A campus coffee cart buys croissants each morning for $1.50 and sells them for $4.00. Whatever is left at closing is sold to the staff for $0.50 each. From its records, daily demand is 40, 60, 80, or 100 croissants with probabilities 20%, 40%, 30%, and 10%. How many should it buy?

alternative	payoff if demand 40	payoff if demand 60	payoff if demand 80	payoff if demand 100	EMV	SD	CV
stock 40	$100.00	$100.00	$100.00	$100.00	$100.00	$0.00	0.000
stock 60	$80.00	$150.00	$150.00	$150.00	$136.00	$28.00	0.206
stock 80	$60.00	$130.00	$200.00	$200.00	$144.00	$52.38	0.364
stock 100	$40.00	$110.00	$180.00	$250.00	$131.00	$63.00	0.481

Each row is an alternative and each payoff column is a state of the world. Stocking 40 earns the same $100 no matter what, because the cart sells out in every state. The highest EMV belongs to **stock 80**, at $144.00 a day.

EMV picks the alternative with the best long-run average. It says nothing about how rough the ride is. Read the SD column: stock 80 averages $44.00 a day more than stocking 40, and in exchange its daily profit swings with a standard deviation of $52.38 where the cautious choice has none.

For a decision repeated every morning, take the higher EMV. The swings wash out, which is exactly what "long-run average" means. For a decision made once, with a downside you could not absorb, EMV alone is not enough.

Insurance is the clean example. A firm can insure a $250,000 piece of equipment against a 2% chance of loss for a $6,000 premium. Going without has an EMV of -$5,000; buying has an EMV of -$6,000. Insurance has the *worse* EMV, which is how insurers stay in business, and buying it can still be the right decision.

The Binomial Distribution

The characteristics of a binomial experiment

A **binomial experiment** has the following characteristics:

1. The experiment consists of a fixed number of trials, denoted by n.
2. Each trial has only two possible outcomes, a **success** or a **failure**.
3. The probability of a success, p, and of a failure, $q = 1 - p$, are constant throughout the experiment.
4. Each trial is independent of the other trials in the experiment.

The random variable x is the number of successes in the n trials. It can be any whole number from 0 to n, and never more than n.

"Success" is a label, not a compliment. If you are counting defective parts, loan defaults, or customers who walk out, then a defect, a default, or a walkout is the success. Assign the label to whichever outcome you are counting, and keep p attached to it.

Building the formula from Chapter 4

Corporate Headquarters believes that a customer who enters the store has a probability of 0.4 of spending $100 or more. Three customers walk in. What is the probability that **exactly one** of them spends $100 or more?

Start with one specific way it could happen: the first customer spends and the other two do not. The customers are independent, so the multiplication rule for independent events gives $pqq = (0.4)(0.6)(0.6)$. But the big spender could just as well be the second customer, or the third.

way	arrangement	probability
1	SFF	0.144
2	FSF	0.144
3	FFS	0.144

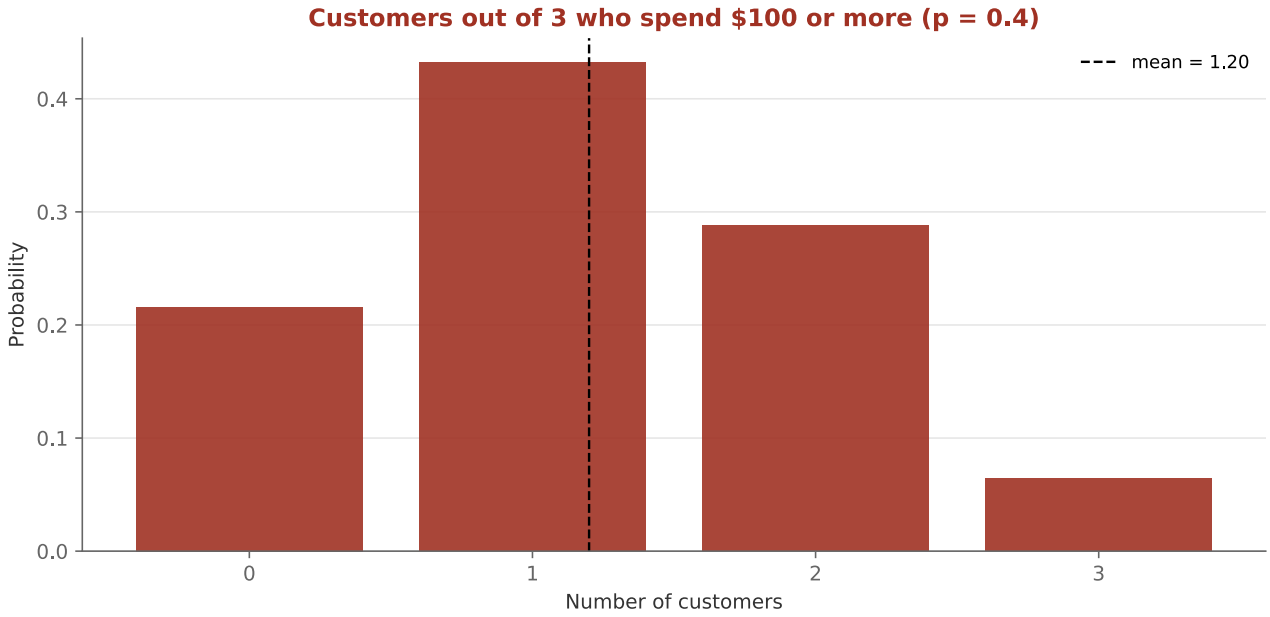
Every arrangement has the same probability, 0.144. The arrangements are mutually exclusive, so the addition rule says to add them:

$$P(x = 1) = 3 \times 0.144 = 0.432$$

That 3 is the number of ways to choose which one of three customers is the spender, ${}_3C_1$, and it is the pattern the end of the Chapter 4 note told you to hold on to. The binomial formula is nothing more than this: **the number of arrangements, times the probability of any one of them.**

> The **binomial probability distribution** gives the probability of exactly x successes in n trials:
>
> $$P(x, n) = \underbrace{\frac{n!}{(n-x)!\,x!}}_{\text{arrangements}} \underbrace{p^x q^{n-x}}_{\text{probability of each}},$$
>
> where p is the probability of a success and $q = 1 - p$ is the probability of a failure.

x	P(X = x)	P(X <= x)	P(X >= x)
0	0.216	0.216	1.000
1	0.432	0.648	0.784
2	0.288	0.936	0.352
3	0.064	1.000	0.064



The `P(X = x)` column sums to 1, as the rules for a distribution require. The other two columns are running totals: the probability of x *or fewer*, and of x *or more*. Most of this chapter's questions are answered from one of those two columns, not from the first.

The mean and standard deviation of a binomial distribution

> For a binomial distribution the general formulas collapse to
>
> $$\mu = np \qquad \text{and} \qquad \sigma = \sqrt{npq}.$$

The mean is the long-run average number of successes in n trials. The standard deviation is how much that count varies from one set of n trials to the next, assuming the chance of success really is p.

Application: Corporate Headquarters

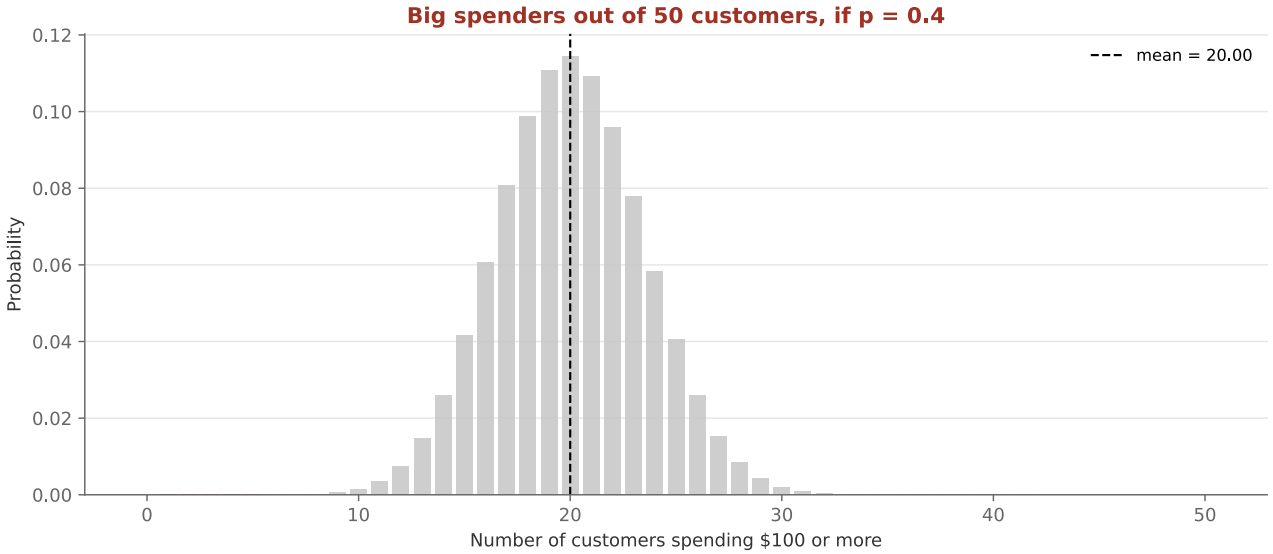
> Corporate Headquarters believes $p = 0.4$. Yesterday 50 customers entered the store and exactly 5 spent $100 or more. According to the binomial distribution, how likely is it that 5 or fewer would?

x	P(X = x)	P(X <= x)	P(X >= x)
0	0.0000000000	0.0000000000	1.0000000000
1	0.0000000003	0.0000000003	1.0000000000
2	0.0000000044	0.0000000047	0.9999999997
3	0.0000000469	0.0000000516	0.9999999953
4	0.0000003677	0.0000004193	0.9999999484
5	0.0000022552	0.0000026745	0.9999995807
6	0.0000112761	0.0000139506	0.9999973255
7	0.0000472521	0.0000612027	0.9999860494
8	0.0001693201	0.0002305229	0.9999387973

$$P(x = 5, n = 50) = \frac{50!}{45!\,5!}(0.4)^5(0.6)^{45} = 2,118,760\,(0.4)^5(0.6)^{45} = 0.000002255$$

$$P(x \le 5) = \sum_{i=0}^{5} \frac{50!}{(50-i)!\,i!}(0.4)^i(0.6)^{50-i} = 0.000002675$$

The probability of exactly 5 is 0.0000023, and of 5 or fewer is 0.0000027, or about 1 in 370,000.



The event we observed is in that chart. It is the scarlet sliver at the far left that you cannot see.

Now interpret it. If p really is 0.4, we expect $\mu = np = 20$ big spenders out of 50, give or take $\sigma = \sqrt{npq} = 3.46$. We saw 5. In the z-score language of Chapter 3 that is 4.3 standard deviations below the mean.

There are two explanations. Either yesterday was a one-in-370,000 day, or Headquarters is wrong about this store and p is not 0.4. A sensible manager does not believe in one-in-370,000 days. She concludes the claim is wrong.

That argument, *assume the claim, compute how surprising the data would be, and reject the claim if the answer is "very"*, is a hypothesis test. We have just done one. Chapter 9 adds the vocabulary and a rule for how surprising is surprising enough.

Notice which probability we used. Not the probability of *exactly* 5, but of *5 or fewer*: the result we saw, together with everything even more damaging to the claim. With 50 trials, every individual outcome is fairly improbable, including the most likely one: P(exactly 20) is only 11.5%. An exact probability being small proves nothing. A *tail* probability being small does.

Translating the question

Most errors in this chapter are made before any calculation, in turning the English into an inequality. A discrete count moves in whole steps, so "fewer than 4" and "4 or fewer" are different events with different probabilities.

the question says	which means	compute	in Excel
exactly 4	X = 4	P(X = 4)	cumulative = FALSE
at most 4; no more than 4; 4 or fewer	X <= 4	P(X <= 4)	cumulative = TRUE
fewer than 4; less than 4	X < 4	P(X <= 3)	cumulative = TRUE at 3
at least 4; 4 or more	X >= 4	1 - P(X <= 3)	1 minus cumulative at 3
more than 4	X > 4	1 - P(X <= 4)	1 minus cumulative at 4

Software gives you two things: the probability of exactly x, and the cumulative probability of x or fewer. Everything else is built from the cumulative one and the complement rule, $P(A) = 1 - P(A')$.

Twelve customers enter the store, still with $p = 0.4$:

```
'exactly 4' means X = 4, computed as P(X = 4) = 0.212841
'at most 4' means X <= 4, computed as P(X <= 4) = 0.438178
'fewer than 4' means X < 4, computed as P(X <= 3) = 0.225337
'at least 4' means X >= 4, computed as 1 - P(X <= 3) = 0.774663
'more than 4' means X > 4, computed as 1 - P(X <= 4) = 0.561822
```

"At least 4" is **one minus the cumulative probability at 3**, not at 4. The complement of "4 or more" is "3 or fewer." Getting this off by one is the single most common mistake in the chapter, and the answer it produces looks perfectly reasonable.

A check that catches it: "at least 4" and "fewer than 4" are complements, so the two must sum to 1. Above, $0.774663 + 0.225337 = 1$.

Reading the output

A bank mails a credit-line increase offer to 15 customers. Historically 30% of customers accept. An analyst asks Claude for the binomial distribution with $n = 15$ and $p = 0.30$ and receives the table below. Using only the table, find the probability that:

(a) exactly 5 customers accept

(b) at most 5 accept

(c) fewer than 5 accept

(d) at least 5 accept

(e) more than 5 accept

(f) between 3 and 6 accept, inclusive

x	P(X = x)	P(X <= x)
0	0.0047	0.0047
1	0.0305	0.0353
2	0.0916	0.1268
3	0.1700	0.2969
4	0.2186	0.5155
5	0.2061	0.7216
6	0.1472	0.8689
7	0.0811	0.9500
8	0.0348	0.9848
9	0.0116	0.9963

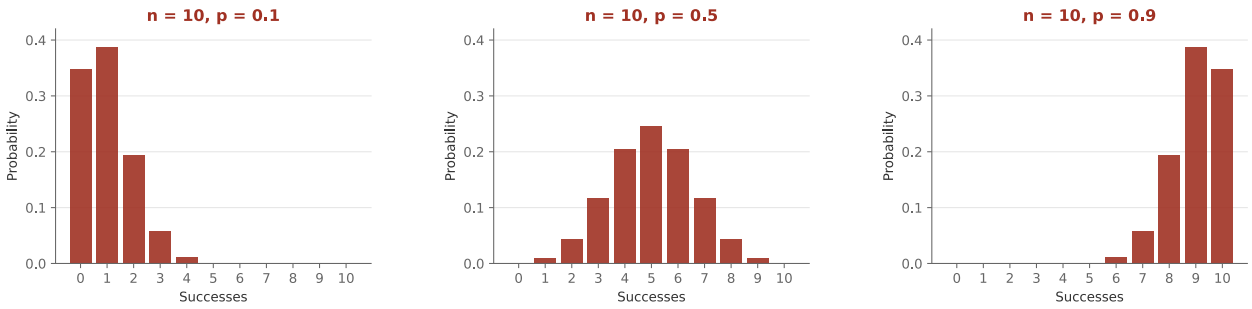
i Solution

part	means	read from the table as	probability
a	X = 5	P(X = 5)	0.2061
b	X <= 5	P(X <= 5)	0.7216
c	X <= 4	P(X <= 4)	0.5155
d	X >= 5	1 - P(X <= 4)	0.4845
e	X > 5	1 - P(X <= 5)	0.2784
f	3 <= X <= 6	P(X <= 6) - P(X <= 2)	0.7420

Part (f) is the one to study. “Between 3 and 6 inclusive” keeps 3, so you subtract the cumulative probability at **2**, the last value you do *not* want. Subtracting the value at 3 would throw away P(X = 3).

Parts (c) and (d) sum to 1, and so do (b) and (e). Run that check every time.

The shape of a binomial distribution



With p below 0.5 the distribution is skewed right, with p above 0.5 it is skewed left, and at $p = 0.5$ it is symmetric. Look back at the chart for 50 customers: even with $p = 0.4$ it is close to symmetric and bell-shaped. As n grows, the binomial approaches the normal curve of Chapter 3, and Chapter 6 puts that to work.

Does the binomial describe Nevada?

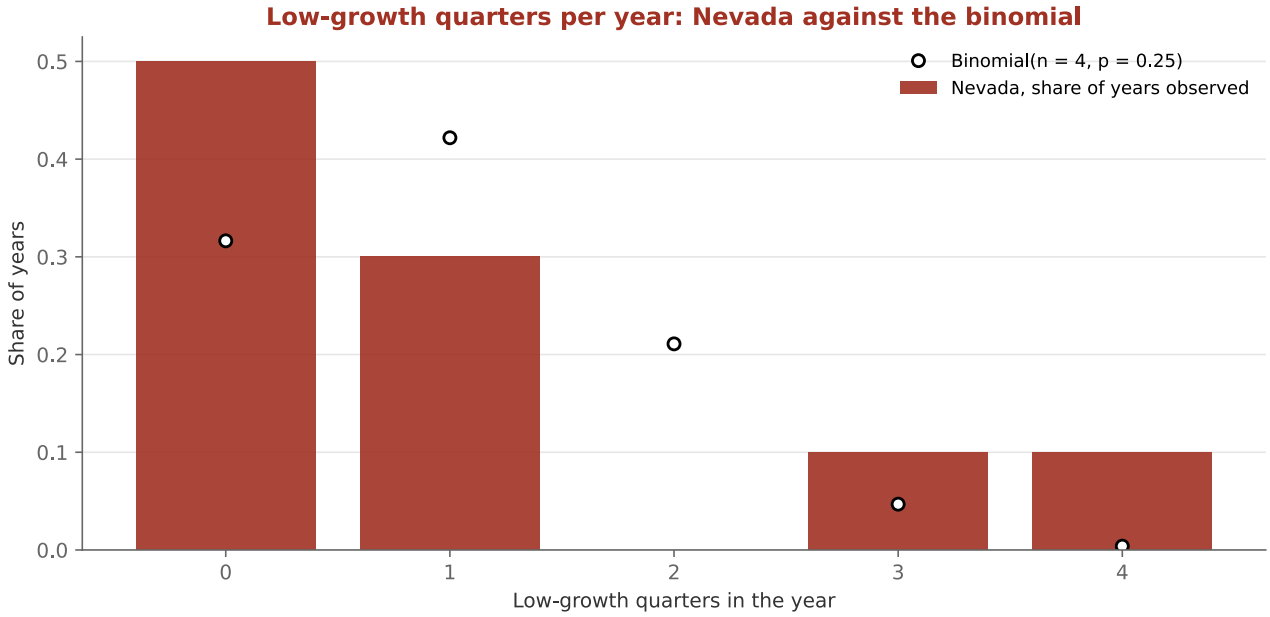
The four characteristics of a binomial experiment are assumptions, and assumptions can be checked. In Chapter 4 we classified each quarter of Nevada’s economy as Low, Middle, or High growth, with Low defined as the bottom 25% of quarters. So for any one quarter, $p = P(\text{Low}) = 0.25$.

A year is four quarters. Is the number of low-growth quarters in a year a binomial random variable with $n = 4$ and $p = 0.25$? There is a fixed number of trials, two outcomes per trial, and a constant p. Three of the four characteristics hold by construction. The fourth is independence.

```
80 quarters, 20 of them Low (growth below 0.19%), so p = 0.25
20 complete years; 0 quarters left over

window     2006  2007  2008  2009  2010  2011  2012  2013  2014  2015  2016  2017  2018  2019  2020  2021  2022
2023  2024  2025
successes     0     1     4     4     1     1     3     1     1     0     0     0     0     0     3     1     0
0     0     0
```

x	windows observed	binomial P(X = x)	windows expected
0	10	0.316	6.328
1	6	0.422	8.438
2	0	0.211	4.219
3	2	0.047	0.938
4	2	0.004	0.078



The means agree: 1.00 low quarters a year observed, $np = 1.00$ predicted. Everything else disagrees. The binomial says a year with all four quarters Low has probability 0.0039, so in 20 years we should expect 0.08 of them. Nevada had 2. The binomial expects about 4 years with exactly two Low quarters, and Nevada had 0. The observed variance is 1.80 against a binomial npq of 0.75, a ratio of 2.4.

Bad quarters arrive in clumps. A low-growth quarter makes the next one more likely to be low, which is the dependence we measured in Chapter 4, and it is precisely what characteristic 4 rules out. Nevada's years are mostly all-clear or all-bad, with little in between.

Some of that clumping is built into the measurement, and you should know it. Our growth figure compares each quarter with the same quarter a year earlier, so neighboring quarters share three-quarters of their data and cannot help moving together. The rest is the economy: recessions last longer than a quarter. Either way the lesson is the same. The trials are not independent, and the reason is visible only to someone who asked how the data were built.

The binomial did not fail quietly here because we went looking. It usually does fail quietly. Loan defaults cluster in recessions. Machine failures cluster after a bad batch of parts. Absences cluster in flu season. In each case a binomial model will get the average right and badly understate how often the extreme outcomes occur, and the extreme outcomes are the ones that sink a business. Before using the binomial, ask what would make one trial's outcome informative about the next.

Using software to calculate binomial probabilities

In Excel, `=BINOM.DIST(x, n, p, cumulative)` returns the probability of exactly x successes when `cumulative` is `FALSE`, and of x or fewer when it is `TRUE`. Those are the first two columns of every table above. `=BINOM.DIST(5, 50, 0.4, TRUE)` reproduces the Corporate Headquarters answer.

Your textbook also prints binomial and Poisson tables in Appendix A. They hold the same numbers for a limited menu of n, p, and λ, and we will not use them.

PROMPT

For a binomial distribution with n = 50 and p = 0.4, compute by running code, not from memory, a table of P(X = x) and P(X <= x) for x from 0 to 50. Then tell me P(X <= 5), and separately P(X >= 25). For each one, state the inequality you evaluated and how you got it from the cumulative column.

Asking for the inequality is how you check the translation. If Claude evaluated $P(X > 25)$ when you wanted $P(X \geq 25)$, the stated inequality is where you will see it.

The Poisson Distribution

A binomial experiment counts successes in a fixed number of trials. Some counts have no trials. How many customers arrive in the next hour? There is no n customers who each either arrived or did not. There is only a stretch of time, and a count of what happened in it.

The characteristics of a Poisson process

A **Poisson process** has the following characteristics:

1. The experiment consists of counting the number of occurrences of an event over a period of time, area, distance, or any other type of measurement.
2. The mean of the Poisson distribution has to be the same for each equal interval of measurement.
3. The number of occurrences during one interval is independent of the number of occurrences in any other interval.
4. The intervals that are defined in the Poisson process cannot overlap.

Typical Poisson random variables: customers arriving at a storefront in an hour, calls reaching a help desk in ten minutes, typographical errors on a page, flaws in a square yard of fabric, workplace injuries in a month.

Using Poisson probability distributions

For the **Poisson probability distribution**,

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!},$$

where x is the number of occurrences of interest over the interval, λ is the mean number of occurrences over the interval, $e \approx 2.71828$, and $P(x)$ is the probability of exactly x occurrences.

The **variance** of a Poisson distribution equals its mean:

$$\sigma^2 = \lambda \qquad \text{so} \qquad \sigma = \sqrt{\lambda}.$$

Two things distinguish the Poisson from the binomial.

λ is an average, so it does **not** have to be a whole number. A help desk can average 4.2 calls an hour. The count x must be whole; its mean need not be.

There is **no upper limit** on x. A binomial count cannot exceed n. A Poisson count has no n, so every whole number has some probability, however tiny. That is why "more than 8" cannot be computed by adding up the terms, and has to be done with the complement rule.

And one thing to notice: the whole distribution is set by a single number. Know λ and you know the mean, the variance, and every probability.

Application: Corporate Headquarters, one more hour

Data scientists at Corporate Headquarters anticipate that 4 additional customers would visit the store if it stayed open an additional hour. What is the probability that more than 8 customers visit in that hour, according to the Poisson distribution?

x	P(X = x)	P(X <= x)	P(X >= x)
0	0.01832	0.01832	1.00000
1	0.07326	0.09158	0.98168
2	0.14653	0.23810	0.90842
3	0.19537	0.43347	0.76190
4	0.19537	0.62884	0.56653
5	0.15629	0.78513	0.37116
6	0.10420	0.88933	0.21487
7	0.05954	0.94887	0.11067
8	0.02977	0.97864	0.05113
9	0.01323	0.99187	0.02136
10	0.00529	0.99716	0.00813
11	0.00192	0.99908	0.00284
12	0.00064	0.99973	0.00092

$$
\begin{aligned}
P(x > 8) &= \sum_{i=9}^{\infty} \frac{\lambda^i e^{-\lambda}}{i!} \\
&= 1 - \sum_{i=0}^{8} \frac{4^i e^{-4}}{i!} \\
&= 1 - 0.97864 \\
&= 0.02136
\end{aligned}
$$



There is a 2.14% chance of more than 8 customers. Note that the sum being subtracted starts at $i = 0$. Zero customers is an outcome, with probability $e^{-4} = 0.0183$, and leaving it out is an easy way to be wrong by almost 2 percentage points.

If staffing for the extra hour can comfortably handle 8 customers, this is the probability the plan is overwhelmed. The standard deviation is $\sqrt{4} = 2$ customers, so 8 is two standard deviations above the mean.

Changing the interval

λ belongs to an interval. Change the interval and you must change λ with it, in proportion, before doing anything else. This works because characteristic 2 says the rate is the same in every equal stretch.

The store averages 4 customers an hour.

(a) What is the probability of exactly 4 customers in the next hour?

(b) What is the probability of exactly 2 customers in the next 30 minutes?

(c) What is the probability of no customers at all in the next 15 minutes?

part	interval (minutes)	lambda for that interval	x asked about	P(X = x)
a	60	4.0	4	0.1954
b	30	2.0	2	0.2707
c	15	1.0	0	0.3679

Parts (a) and (b) look as though they ought to match: half the time, half the customers. They do not. Exactly 4 in an hour has probability 0.1954 and exactly 2 in half an hour has probability 0.2707.

You rescale the **mean**. You cannot rescale the **probability**. Whenever the interval in the question differs from the interval the average was quoted for, the first line of your work is a new λ.

Using the Poisson distribution to approximate the binomial

When the number of trials is large and the probability of success is small, specifically when

$$n \geq 20 \qquad \text{and} \qquad p \leq 0.05,$$

binomial probabilities can be approximated with the Poisson formula by setting $\lambda = np$:

$$P(x) \approx \frac{(np)^x e^{-np}}{x!}.$$

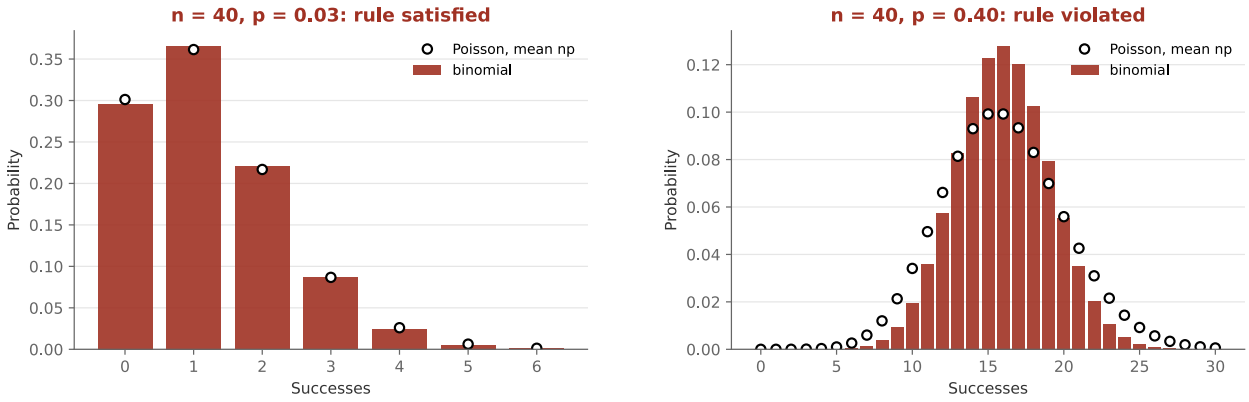
A distribution center mis-packs 3% of its orders. Forty orders are pulled for inspection. What is the probability that exactly 2 are mis-packed?

```
lambda = np = 1.20;  n >= 20 and p <= 0.05: True
```

x	binomial	Poisson	difference
0	0.2957	0.3012	0.0055
1	0.3658	0.3614	−0.0044
2	0.2206	0.2169	−0.0038
3	0.0864	0.0867	0.0003
4	0.0247	0.0260	0.0013
5	0.0055	0.0062	0.0007
6	0.0010	0.0012	0.0003

The exact binomial answer is 0.2206 and the Poisson approximation is 0.2169. No entry in the table is off by more than 0.0055.

The conditions matter. Keep $n = 40$ but let $p = 0.40$, and the two distributions part company:



With $p = 0.40$ the largest gap is 0.0287. The Poisson is too spread out, because its variance is np while the binomial's is the smaller npq. When p is tiny, q is nearly 1 and the two variances are nearly equal, which is the whole reason the approximation works.

A fair question: if software computes the binomial exactly, why approximate it? For arithmetic, you no longer need to. The textbook's motive, fewer calculator keystrokes, has expired.

The idea has not. It explains *why* so many business counts are Poisson. A count of rare events, such as typos, accidents, fraud cases, or warranty claims, is a binomial with an enormous n (every keystroke, every mile driven, every transaction) and a minuscule p. You will never know that n or that p. The approximation says you do not need them. Their product, the average count λ, is enough, and that you can estimate from records.

Using software to calculate Poisson probabilities

In Excel, `=POISSON.DIST(x, lambda, cumulative)` works exactly like `BINOM.DIST`: `FALSE` for exactly x occurrences, `TRUE` for x or fewer. The Corporate Headquarters answer is `=1-POISSON.DIST(8, 4, TRUE)`.

The Hypergeometric Distribution

The **hypergeometric distribution** is used when samples are taken from a small population without being replaced.

The binomial requires a constant p. In Chapter 4 an auditor drew invoices from a batch of 32, of which 9 contained an error, and we saw that the probability of an error on the second draw depended on the first: 9/32, then 8/31 or 9/31. When you sample without replacement from a small population, the probability of a success on each draw depends on the draws before it. The trials are not independent with a constant p, so characteristics 3 and 4 fail together and the binomial does not apply.

For the **hypergeometric distribution**,

$$P(x) = \frac{{}_{N-R}C_{n-x} \cdot {}_{R}C_{x}}{{}_{N}C_{n}},$$

where N is the population size, R is the number of successes in the population, n is the sample size, and x is the number of successes in the sample.

Read it as a classical probability from Chapter 4. The denominator counts every sample of n that could be drawn. The numerator counts the samples with exactly x successes: the ways to choose x from the R successes, times the ways to choose the other $n - x$ from the $N - R$ failures.

The **mean** and **standard deviation** of the hypergeometric distribution are

$$\mu = \frac{nR}{N} \qquad \text{and} \qquad \sigma = \sqrt{\frac{nR(N-R)}{N^2}}\sqrt{\frac{N-n}{N-1}}.$$

The auditor, continued

Suppose the auditor pulls 5 of the 32 invoices. How many will contain an error?

x	hypergeometric (without replacement)	binomial (as if replaced)	difference
0	0.1671	0.1918	0.0247
1	0.3958	0.3753	−0.0205

x	hypergeometric (without replacement)	binomial (as if replaced)	difference
2	0.3166	0.2937	−0.0229
3	0.1055	0.1149	0.0094
4	0.0144	0.0225	0.0081
5	0.0006	0.0018	0.0011

The left column is correct. The right column is what the binomial would say if we pretended each invoice was put back, with $p = R/N = 0.2812$. Both have the same mean, 1.406 bad invoices. With $p = R/N$, the hypergeometric mean nR/N *is* np.

The standard deviations differ: 0.938 for the hypergeometric against 1.005 for the binomial. Look at the formula again. The first square root is $\sqrt{npq}$. The second, $\sqrt{(N-n)/(N-1)} = 0.933$, is a shrink factor. Sampling without replacement is *less* variable, because every invoice you pull is information you cannot draw again. The factor approaches 1 as N grows, and then the binomial is a fine approximation. It returns in Chapter 7 under the name **finite population correction**.

Application: selecting store managers

A regional vice president would like to select 3 store managers at random from the 12 area stores. Two of the area store managers have less than 2 years of experience. What is the probability that none of the managers with less than 2 years of experience is selected?

i Solution

x	P(X = x)	P(X <= x)	P(X >= x)
0	0.5455	0.5455	1.0000
1	0.4091	0.9545	0.4545
2	0.0455	1.0000	0.0455
3	0.0000	1.0000	0.0000

$$P(x=0) = \frac{{}_{12-2}C_{3-0} \cdot {}_{2}C_{0}}{{}_{12}C_{3}} = \frac{120 \times 1}{220} = 0.5455$$

Of the 220 possible groups of three, 120 contain only experienced managers, so the probability is 54.55%.

Application: was the selection random?

A casino's compliance group audits 6 of its 25 slot-machine cabinets each quarter, chosen, it says, at random. Eight of the 25 cabinets come from one vendor. This quarter, 5 of the 6 audited cabinets were that vendor's, and the vendor has complained of being singled out. Does the vendor have a case?

i Solution

Define a success as selecting one of the vendor's cabinets: $N = 25$, $R = 8$, $n = 6$, and we observed $x = 5$.

x	P(X = x)	P(X <= x)	P(X >= x)
0	0.06988	0.06988	1.00000
1	0.27953	0.34941	0.93012
2	0.37628	0.72569	0.65059
3	0.21502	0.94071	0.27431
4	0.05375	0.99447	0.05929
5	0.00538	0.99984	0.00553
6	0.00016	1.00000	0.00016

Under random selection we would expect $\mu = nR/N = 1.92$ of the vendor's cabinets in the sample, with a standard deviation of 1.02. The probability of 5 *or more* is 0.00553, about 1 in 180.

This is the Corporate Headquarters argument again. Either the casino drew a 1-in-180 sample, or the selection was not random. The vendor has a case worth answering. Calculations of exactly this kind are offered as evidence in discrimination suits over layoffs and promotions, where the "population" is a workforce and the "sample" is who was let go.

As before, we used the tail, "5 or more," not "exactly 5."

Using software to calculate hypergeometric probabilities

In Excel, `=HYPGEOM.DIST(x, n, R, N, cumulative)`, where x is the number of successes in the sample, n is the sample size, R is the number of successes in the population, and N is the population size. `=HYPGEOM.DIST(0, 3, 2, 12, FALSE)` reproduces the store managers answer. The helper functions in this note take their arguments in the same order.

Which Distribution Is It?

On an exam and in practice, nobody tells you which distribution applies. Three questions settle it.

Ask	If yes
Is there a fixed number of trials, each a success or failure, with constant p and independent trials?	**Binomial**, with n and p
Same, but sampling *without replacement* from a *small* population, so p shifts with each draw?	**Hypergeometric**, with N, R, n
No trials at all, just a count of occurrences over an interval of time or space, with a known average?	**Poisson**, with λ

Name the distribution and its parameters for each of the following.

(a) An airline finds 7% of ticketed passengers do not show up. It sells 60 tickets for a flight. How many no-shows?

(b) A hotel front desk averages 9 check-ins per hour on Friday evenings. How many in the next 20 minutes?

(c) A department of 14 people includes 5 who hold a PMP certification. Four are picked at random for a project team. How many on the team are certified?

(d) A survey firm phones 200 households drawn from the 900,000 households in Clark County, where 31% own a pet. How many pet owners in the sample?

(e) A textile mill averages 1.8 flaws per 100 yards of fabric. How many flaws in a 250-yard roll?

(f) Each of 18 sales calls ends in a sale or not. The rep closes 25% of calls in general, but tends to go on hot and cold streaks. How many sales?

i Solution

(a) Binomial, $n = 60$, $p = 0.07$. The Poisson shortcut is not licensed, because p is above 0.05, though with n this large it would not be far off.

(b) Poisson. The average is quoted per hour and the question is about 20 minutes, so $\lambda = 9 \times \frac{20}{60} = 3$.

(c) Hypergeometric, $N = 14$, $R = 5$, $n = 4$. The sample is 29% of the population and is drawn without replacement.

(d) Strictly hypergeometric, since nobody is phoned twice. In practice binomial with $n = 200, p = 0.31$: the sample is 0.022% of the population, and the shrink factor $\sqrt{(N-n)/(N-1)}$ is 0.99989. This is why polls are analysed with the binomial.

(e) Poisson, with $\lambda = 1.8 \times \frac{250}{100} = 4.5$. The interval is a length, not a time.

(f) It has the form of a binomial with $n = 18$ and $p = 0.25$, and streaks are a violation of independence. This is Nevada's economy again. A binomial will get the rep's average right and understate how often a very good or very bad week occurs.

Applications

Attempt each one before opening the solution.

The help desk

An IT help desk averages 18 calls per hour. Its two technicians can clear 5 calls in any 10-minute window without a queue forming. Assume calls follow a Poisson distribution.

(a) What is the probability that a queue forms in a given 10-minute window?

(b) What is the probability that a 5-minute coffee break goes uninterrupted?

(c) What assumption is most likely to be wrong, and which way does it push the answer to (a)?

Solution

```
(a) lambda for 10 minutes = 3.0
    'more than 5' means X > 5, computed as 1 - P(X <= 5) = 0.083918
(b) lambda for 5 minutes = 1.5;  P(X = 0) = 0.2231
```

(a) A queue forms when *more than* 5 calls arrive, so we need $P(X > 5) = 1 - P(X \leq 5)$ with $\lambda = 3$: 8.39%. About one window in 12.

(b) 22.31%. Better than even odds the break is interrupted.

(c) Characteristic 2, a constant rate. Help desk calls are not spread evenly through the day; they pile up at 9 a.m. and after lunch. An average of 18 an hour conceals busy hours well above 18 and quiet hours well below. During the busy hours the chance of a queue is far higher than 8%, and that is when it matters. Characteristic 3 is suspect too: one system outage produces a burst of calls that are anything but independent.

The supplier's claim

A supplier guarantees that no more than 5% of its components are defective. Your receiving dock tests 20 components from a very large shipment and finds 4 defective.

(a) If the supplier's claim is exactly true, how many defectives would you expect, and with what standard deviation?

(b) What is the probability of finding 4 or more?

(c) What do you conclude? What would you need to believe for the calculation to be valid?

i Solution

```
(a) mean = np = 1.00,  sd = sqrt(npq) = 0.975
(b) 'at least 4' means X >= 4, computed as 1 - P(X <= 3) = 0.015902
```

x	P(X = x)	P(X <= x)	P(X >= x)
0	0.35849	0.35849	1.00000
1	0.37735	0.73584	0.64151
2	0.18868	0.92452	0.26416
3	0.05958	0.98410	0.07548
4	0.01333	0.99743	0.01590
5	0.00224	0.99967	0.00257
6	0.00030	0.99997	0.00033

(a) We expect 1 defective, give or take 0.97.

(b) $P(X \geq 4) = 1 - P(X \leq 3) = 0.0159$.

(c) If the claim were true, a sample this bad would turn up about 1.6% of the time, roughly once in 63 shipments. That is strong evidence that the defect rate in this shipment is above 5%, and grounds to hold the shipment and call the supplier.

It is not proof. One shipment in 63 from an honest supplier would look like this. How small a probability has to be before you act is a business judgment about the cost of being wrong in each direction, and Chapter 9 gives that judgment a name.

For the calculation to hold, the 20 components must be a random sample (not the top layer of one pallet), and one component's being defective must not make its neighbor's more likely. If defects come in runs from a single bad production hour, the trials are dependent, and as with Nevada the binomial would understate the chance of a cluster like this one. The shipment being "very large" is what lets us use the binomial instead of the hypergeometric.

Laboratory 5

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 5 pack — one file. Unzip it and you will have a folder holding 6 files. Drag all 6 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

> I have uploaded 6 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.

> Then run `05_discrete_distributions.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

> **PROMPT**
>
> Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 5 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.
>
> Then run `05_discrete_distributions.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.
>
> If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach the page, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Your own junior analyst

The lab script answers the questions in this note. The skill to practice is getting a *new* question answered and checking the answer. Try this, with your own numbers in place of mine:

> **PROMPT**
>
> A call center averages 42 calls per hour. Assume the Poisson distribution. By running code, find the probability of more than 10 calls in the next 10 minutes. Before you compute anything, tell me (1) the value of lambda you are using and why, and (2) the inequality you are evaluating and how you will get it from a cumulative probability. Then show the code and its output.

Check three things in the reply: that λ was rescaled to 10 minutes, that "more than 10" became $1 - P(X \leq 10)$ and not $1 - P(X \leq 9)$, and that the number came from executed code. Those are the three places this chapter goes wrong, and you now know enough to catch all three.

Working in Excel

> **PROMPT**
>
> Build me an Excel workbook with three sheets, one each for the binomial, Poisson, and hypergeometric distributions. On each sheet, the parameters (n and p; lambda; or N, R and n) are editable input cells at the top. Below them, build a table of x, P(X = x), P(X <= x), and P(X >= x) using BINOM.DIST, POISSON.DIST, and HYPGEOM.DIST, with real formulas that reference the input cells. Add cells for the mean and standard deviation of each distribution, and a column chart of P(X = x). Pre-fill the binomial sheet with n = 50 and p = 0.4 from my Chapter 5 lecture notes. Give me the file to download.

Change p on the binomial sheet from 0.4 to 0.1 and watch the chart slide left and skew. Then set $n = 40$ and $p = 0.03$ on the binomial sheet and $\lambda = 1.2$ on the Poisson sheet, and put the two `P(X = x)` columns side by side. Answer in one paragraph: at what value of p, holding $n = 40$, would *you* stop trusting the Poisson approximation, and what did you look at to decide?